

Day 96 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in General Science (8): India's Renewable Energy Capacity Milestones (2026)

READING MATERIAL

Covers India's renewable energy capacity milestones as of early-to-mid 2026.

Q: What is India's total renewable energy capacity as of 31 March 2026, and what is its global ranking?

A: 288 GW — making India the world's THIRD-LARGEST renewable energy nation.

Q: How is India's renewable energy capacity split by source (as of 31 March 2026)?

A: Solar ~55%, Wind ~20%, Large Hydro ~19%, Biomass ~4%, Small Hydro ~2%.

Q: What solar capacity milestone did India cross by 31 March 2026, and how did it grow by end-May 2026?

A: 150.26 GW by 31 March 2026; grew further to approximately 157.05 GW by end of May 2026.

Q: How much solar capacity did India add in FY2026, and what was the year-on-year growth rate?

A: About 44.6 GW added in FY2026 — an 87.2% year-on-year increase in solar installations.

Q: How has India's domestic solar MODULE manufacturing capacity grown, as of 31 March 2026?

A: From 2.3 GW to 172 GW — a dramatic increase in domestic manufacturing capability, not just installed capacity.

MOCK TEST (WITH ANSWERS)

1. As of 31 March 2026, India's total renewable energy capacity (288 GW) ranks it as the world's:

- A. Largest
- B. 2nd largest
- C. 3rd largest ✓**
- D. 5th largest

Explanation: 3rd largest renewable energy nation. [medium, web-research]

2. Which source contributes the largest share of India's renewable energy capacity (as of 31 March 2026)?

- A. Wind
- B. Solar ✓**
- C. Large Hydro
- D. Biomass

Explanation: Solar (~55%). [easy, web-research]

3. India's cumulative installed solar capacity crossed 150 GW by:

- A. 31 December 2025
- B. 31 January 2026
- C. 31 March 2026 ✓**
- D. 31 May 2026

Explanation: 31 March 2026 (150.26 GW). [medium, web-research]

4. India added approximately how much solar capacity in FY2026?

- A. 24.6 GW
- B. 34.6 GW
- C. 44.6 GW ✓**
- D. 54.6 GW

Explanation: 44.6 GW (87.2% YoY growth). [hard, web-research]

5. India's domestic solar module manufacturing capacity grew from 2.3 GW to approximately how much (as of 31 March 2026)?

- A. 72 GW
- B. 122 GW

C. 172 GW ✓

D. 222 GW

Explanation: 172 GW. [hard, web-research]

Part B – Aptitude & Mental Ability: Number Series

READING MATERIAL

Number Series is the final Reasoning skill. Real exam evidence shows 6 distinct underlying patterns: second-difference (difference-of-differences) sequences, sum-of-consecutive-cubes, geometric progression, arithmetic progression, n^2 multiples, and prime-number-squared sequences — recognize the pattern TYPE before trying to compute term-by-term.

Q: Method — sum of consecutive cubes: Find the next term: 9, 35, 91, ?

A: Each term is $n^3 + (n+1)^3$: $1^3 + 2^3 = 1 + 8 = 9$ ✓, $2^3 + 3^3 = 8 + 27 = 35$ ✓, $3^3 + 4^3 = 27 + 64 = 91$ ✓. Next: $4^3 + 5^3 = 64 + 125 = 189$.

Q: Method — second-difference (arithmetic sequence of differences): Fill in the blank: 1, 6, 15, 28, 45, ____, 91.

A: Differences: $6 - 1 = 5$, $15 - 6 = 9$, $28 - 15 = 13$, $45 - 28 = 17$ — these differences themselves increase by 4 each time. Next difference = $17 + 4 = 21$, so the blank = $45 + 21 = 66$. (Check: $91 - 66 = 25 = 21 + 4$ ✓ consistent.)

Q: Method — sequence of squared primes: Find the missing number in the sequence of squared prime numbers: 4 (2^2), 9 (3^2), 25 (5^2), 49 (7^2), ?, 169 (13^2).

A: The base numbers are consecutive primes: 2, 3, 5, 7, 11, 13. The missing term corresponds to $11^2 = 121$. (Caution: the original source rendered this sequence with garbled formatting that looked like plain 3-digit numbers rather than squared primes — always re-derive the actual pattern rather than trusting a confusing transcription.)

Q: Method — simple arithmetic progression (AP) with increasing-by-2 differences: Find the next term: 1, 5, 11, 19, 29, ?

A: Differences: 4, 6, 8, 10 — increasing by 2 each time. Next difference = 12, so next term = $29 + 12 = 41$.

Q: Method — geometric progression (GP): Find the 8th term of 2, 6, 18, 54, ...

A: Common ratio = 3 (each term $\times 3$). Terms: 2, 6, 18, 54, 162, 486, 1458, 4374. The 8th term is 4,374.

Q: Method — plain arithmetic progression, nth term formula: Find the 20th term of 7, 10, 13, ...

A: First term = 7, common difference = 3. nth term = $a + (n-1)d = 7 + (19 \times 3) = 7 + 57 = 64$.

Q: Method — multiples of n^2 : Find the missing number: 3, 12, 27, 48, 75, 108, ____.

A: Each term = $3 \times n^2$ for $n=1, 2, 3, \dots$: $3 \times 1 = 3$, $3 \times 4 = 12$, $3 \times 9 = 27$, $3 \times 16 = 48$, $3 \times 25 = 75$, $3 \times 36 = 108$. Next ($n=7$): $3 \times 49 = 147$.

MOCK TEST (WITH ANSWERS)

1. Find the next term: 2, 9, 28, 65, 126, ?

A. 215

B. 217 ✓

C. 219

D. 221

Explanation: Each term is $n^3 + 1$: $1^3 + 1 = 2$, $2^3 + 1 = 9$, $3^3 + 1 = 28$, $4^3 + 1 = 65$, $5^3 + 1 = 126$. Next: $6^3 + 1 = 216 + 1 = 217$. [hard, authored]

2. Fill in the blank: 2, 7, 14, 23, 34, ____, 62.

A. 45

B. 47 ✓

C. 49

D. 51

Explanation: Differences: 5, 7, 9, 11 — increasing by 2. Next difference = 13, so blank = $34 + 13 = 47$. (Check: $62 - 47 = 15 = 13 + 2$, consistent.) [medium, authored]

3. Find the next term: 3, 7, 15, 27, 43, ?

A. 61

B. 63 ✓

C. 65

D. 67

Explanation: Differences: 4, 8, 12, 16 — increasing by 4. Next difference = 20, so next term = $43 + 20 = 63$. [medium, authored]

4. Find the 7th term of the GP: 3, 9, 27, 81, ...

A. 1,458

B. 2,187 ✓

C. 2,287

D. 1,987

Explanation: Common ratio 3: 3,9,27,81,243,729,2187. 7th term = 2,187. [medium, authored]

5. Find the 15th term of the AP: 5, 9, 13, 17, ...

A. 59

B. 61 ✓

C. 63

D. 65

Explanation: $a=5$, $d=4$. 15th term = $5+(14 \times 4) = 5+56 = 61$. [easy, authored]

6. Find the missing number: 2, 8, 18, 32, 50, ____.

A. 62

B. 68

C. 72 ✓

D. 78

Explanation: Each term = $2 \times n^2$: $2 \times 1=2$, $2 \times 4=8$, $2 \times 9=18$, $2 \times 16=32$, $2 \times 25=50$. Next ($n=6$): $2 \times 36=72$. [medium, authored]

Part C – General English: Poem: "A Poison Tree" — William Blake

READING MATERIAL

A dark, allegorical poem about the danger of suppressed anger, where unexpressed wrath grows like a tree bearing poisoned fruit. Real coaching-site evidence tests figure-of-speech identification per line, heavily.

Q: What is the poem's plot and central theme?

A: The poet tells his friend about his anger, and the anger ends. But he hides his anger from his enemy instead — and that suppressed wrath grows, like a tended tree, eventually bearing a 'poisonous apple.' His enemy steals into the garden at night, eats the apple, and is found dead under the tree the next morning. Central theme: suppressed anger/enmity is far more dangerous than openly expressed anger — 'enmity dies hard' when concealed rather than resolved.

Q: What is the central, extended metaphor of the poem?

A: Anger is compared to a tree that the poet 'grows' — nurtured with 'fears' (as water) and 'tears' (day and night), eventually bearing a deadly fruit. The poet's relationship to his suppressed wrath is like a gardener's relationship to a plant he deliberately cultivates.

Q: Identify the figure of speech: "I was angry with my friend; I told my wrath, my wrath did end."

A: Anaphora — repetition of 'my wrath' across the line for emphasis.

Q: Identify the figure of speech: "And I sunned it with smiles, / And with soft deceitful wiles."

A: Alliteration — repeated 's' and 'w' sounds ('sunned'/'smiles'; 'soft'/'deceitful'/'wiles').

Q: Identify the figure of speech: "Till it bore an apple bright" and "And I watered it in fears, / Night & morning with my tears."

A: Both are Metaphor — the suppressed anger is directly described as a tree bearing fruit and being watered, without using 'like/as'.

Q: Identify the figure of speech: "When the night had veiled the pole."

A: Personification — night is given the human-like action of 'veiling' (covering, as with a veil).

Q: What is the rhyme scheme of the poem?

A: AABB — rhyming couplets.

MOCK TEST (WITH ANSWERS)

1. What happened when the poet told his FRIEND about his anger?

- A. The anger grew worse
- B. The anger ended ✓**
- C. The friend got angry too
- D. Nothing changed

Explanation: Telling his friend, the wrath ended — contrasted with hiding it from his enemy. [easy, web-research]

2. What does the 'poison tree' in the poem represent?

- A. A real tree in the poet's garden
- B. Suppressed/concealed anger ✓**
- C. Friendship
- D. Forgiveness

Explanation: Suppressed anger, nurtured in secret until it becomes deadly. [easy, web-research]

3. What happens to the enemy at the end of the poem?

- A. He apologizes to the poet
- B. He is found dead under the tree, having eaten the poisoned apple ✓**
- C. He destroys the tree
- D. He becomes the poet's friend

Explanation: He sneaks in at night, eats the apple, and is found dead under the tree. [medium, web-research]

4. "I was angry with my friend; I told my wrath, my wrath did end" is an example of:

- A. Alliteration
- B. Anaphora ✓**
- C. Metaphor
- D. Personification

Explanation: Anaphora — repetition of 'my wrath'. [medium, web-research]

5. "Till it bore an apple bright" uses which figure of speech?

A. Simile

B. Metaphor ✓

C. Hyperbole

D. Personification

Explanation: Metaphor — anger directly described as a fruit-bearing tree. [medium, web-research]

6. "When the night had veiled the pole" uses which figure of speech?

A. Hyperbole

B. Alliteration

C. Personification ✓

D. Metaphor

Explanation: Personification — night given the human action of 'veiling'. [medium, web-research]

7. "A Poison Tree" follows which rhyme scheme?

A. ABAB

B. AABB ✓

C. ABBA

D. No rhyme

Explanation: AABB, rhyming couplets. [medium, web-research]